Light

GCSE Computer Science — CPU Architecture (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

In the Fetch-Decode-Execute cycle, what happens during the 'fetch' stage?

- ○ **A.** The CPU works out what the instruction means
- ○ **B.** The instruction is carried out by the appropriate component
- ○ **C.** The next instruction is retrieved from main memory
- ○ **D.** The result of the instruction is written back to memory

Q2 (1 mark)

What occurs during the 'decode' stage of the Fetch-Decode-Execute cycle?

- ○ **A.** The instruction is interpreted to determine what action is required
- ○ **B.** Data is transferred from RAM to the CPU registers
- ○ **C.** The program counter is reset to zero
- ○ **D.** The ALU performs the required calculation

Q3 (1 mark)

During which stage of the Fetch-Decode-Execute cycle would an addition operation actually be performed?

- ○ **A.** Fetch
- ○ **B.** Decode
- ○ **C.** Store
- ○ **D.** Execute

Q4 (1 mark)

Which CPU component is responsible for decoding instructions?

- ○ **A.** The arithmetic logic unit
- ○ **B.** The control unit
- ○ **C.** The system clock
- ○ **D.** The data register

Q5 (1 mark)

Which type of memory contains instructions that are permanently stored and cannot be modified during normal operation?

- ○ **A.** RAM
- ○ **B.** Cache
- ○ **C.** Register
- ○ **D.** ROM

Q6 (1 mark)

What does the term 'volatile' mean when describing computer memory?

- ○ **A.** The memory can be easily damaged by static electricity
- ○ **B.** The memory contents are lost when power is removed
- ○ **C.** The memory can only store data for a limited time before it expires
- ○ **D.** The memory operates at very high temperatures

Q7 (1 mark)

What is 'main memory' in a computer system?

- ○ **A.** The hard drive where files are permanently stored
- ○ **B.** The graphics card memory used for displaying images
- ○ **C.** Memory directly accessible by the CPU, typically RAM

- ○ **D.** External USB storage devices

Q8 (1 mark)

A CPU operates at 2.8 GHz. What does this measurement indicate?

- ○ **A.** The CPU clock generates 2.8 billion pulses per second
- ○ **B.** The CPU can process 2.8 gigabytes of data per second
- ○ **C.** The CPU contains 2.8 billion transistors
- ○ **D.** The CPU has 2.8 billion bytes of cache memory

Q9 (1 mark)

What is the main advantage of having multiple cores in a CPU?

- ○ **A.** Each core can store more data than a single core
- ○ **B.** Multiple cores allow the computer to use more RAM
- ○ **C.** Each core increases the clock speed of the processor
- ○ **D.** Multiple cores can process different tasks simultaneously

Q10 (1 mark)

In a modern CPU, which level of cache is typically the smallest but fastest?

- ○ **A.** L1 cache

- B. L2 cache
- C. L3 cache
- D. L4 cache

Q11 (1 mark)

Which bus allows data to flow in both directions between the CPU and memory?

- A. Address bus
- B. Control bus
- C. Data bus
- D. Power bus

Q12 (1 mark)

Why are registers used in a CPU instead of accessing data directly from RAM for every operation?

- A. Registers can hold larger amounts of data than RAM
- B. Registers are less expensive to manufacture than RAM
- C. Registers are non-volatile while RAM is volatile
- D. Registers provide much faster access than RAM

Q13 (2 marks)

A program instruction tells the CPU to add two numbers stored in memory locations A and B, then store the result in location C. Which sequence correctly describes the CPU's actions?

- ○ **A.** Execute the addition, then fetch the instruction, then decode the result
- ○ **B.** Fetch the instruction, decode it to understand the addition is needed, execute by performing the calculation and storing the result
- ○ **C.** Decode the memory locations, fetch the numbers, execute the storage
- ○ **D.** Fetch both numbers simultaneously, decode them into binary, execute the addition

Q14 (2 marks)

A computer technician explains that the BIOS must be stored in a specific type of memory. Why is ROM used for BIOS storage instead of RAM?

- ○ **A.** ROM is faster than RAM, so the computer starts up more quickly
- ○ **B.** ROM has larger storage capacity than RAM, allowing more complex BIOS programs
- ○ **C.** ROM retains its contents without power, so the BIOS is available when the computer is first switched on
- ○ **D.** ROM is cheaper than RAM, reducing the manufacturing cost of motherboards

Q15 (2 marks)

When a CPU requests data that is not found in the cache, this is called a 'cache miss'. What happens after a cache miss occurs?

- ○ **A.** The CPU stops processing and waits for the user to provide the data
- ○ **B.** The CPU generates the missing data using calculations in the ALU
- ○ **C.** The operating system automatically terminates the program
- ○ **D.** The data must be retrieved from the slower main memory (RAM)

Q16 (2 marks)

A user increases their CPU's clock speed from 3.5 GHz to 4.0 GHz through overclocking. Which statement correctly describes a potential consequence?

- ○ **A.** The CPU will automatically gain additional cores to handle the extra speed
- ○ **B.** The CPU may generate more heat and require better cooling
- ○ **C.** The amount of RAM in the system will effectively increase
- ○ **D.** The CPU's cache size will automatically expand to match the new speed

Q17 (2 marks)

A computer system has a 64-bit data bus. Compared to a 32-bit data bus, what is the main advantage?

- ○ **A.** The 64-bit bus operates at twice the clock speed
- ○ **B.** The 64-bit bus consumes half the power
- ○ **C.** The 64-bit bus can address twice as many memory locations

○ **D.** The 64-bit bus can transfer twice as much data in each operation

Q18 (2 marks)

The control unit needs to tell the memory to read data from a specific location. Which combination of buses does it use?

○ **A.** Data bus only — to send the read request and receive the data

○ **B.** Address bus and data bus only — to specify the location and transfer the data

○ **C.** Control bus only — to manage all aspects of the memory operation

○ **D.** Address bus to specify the location, control bus to signal a read operation, data bus to receive the data

Q19 (2 marks)

The Fetch-Decode-Execute cycle is described as 'continuous'. What does this mean for how the CPU operates?

○ **A.** The CPU repeats the cycle endlessly, processing one instruction after another until the computer is powered off or halted

○ **B.** The CPU completes one cycle and then waits for user input before continuing

○ **C.** The CPU runs the cycle once when the computer starts and once when it shuts down

○ **D.** The CPU performs all three stages simultaneously for maximum efficiency

Q20 (2 marks)

During a subtraction operation, what are the respective roles of the control unit and the ALU?

- ○ **A.** The control unit performs the subtraction while the ALU stores the result
- ○ **B.** The control unit directs the operation and the ALU performs the calculation
- ○ **C.** The control unit stores the numbers while the ALU decodes the instruction
- ○ **D.** The control unit and ALU both perform identical subtraction operations for verification

Q21 (2 marks)

Arrange the following memory types in order from fastest to slowest access time: RAM, Register, Hard Drive, Cache.

- ○ **A.** Hard Drive, RAM, Cache, Register
- ○ **B.** Register, Cache, RAM, Hard Drive
- ○ **C.** Cache, Register, RAM, Hard Drive
- ○ **D.** RAM, Register, Cache, Hard Drive

Q22 (2 marks)

A student claims that doubling the clock speed will double the computer's overall performance. Why is this statement not entirely accurate?

- ○ **A.** Clock speed is measured in bytes, not operations per second

○ **B.** Doubling clock speed quadruples the power consumption, limiting actual performance

○ **C.** Performance depends on multiple factors including memory speed, which may become a bottleneck

○ **D.** The operating system cannot handle clock speeds above a certain threshold

Q23 (2 marks)

During the execution of a program, the CPU needs to temporarily hold the result of a calculation before using it in the next instruction. Where is this intermediate result most likely stored?

○ **A.** In secondary storage for long-term retention

○ **B.** In ROM where it cannot be accidentally modified

○ **C.** In the control unit's instruction decoder

○ **D.** In a CPU register for immediate access

Q24 (3 marks)

A database server processes thousands of queries simultaneously. The IT manager is considering two CPU options:

- **CPU Alpha:** 12 cores at 3.2 GHz, 30 MB cache
- **CPU Beta:** 6 cores at 4.5 GHz, 18 MB cache

The database software is highly optimised for multi-threading. Which CPU would better suit this workload, and why?

○ **A.** CPU Beta, because the higher clock speed means each query will be processed faster

○ **B.** CPU Beta, because fewer cores means less coordination overhead between processing units

○ **C.** CPU Alpha, because more cores allow more queries to be processed simultaneously, and larger cache reduces database access times

○ **D.** Both would perform identically since the total computational power is similar

Q25 (3 marks)

A simple CPU without pipelining takes 3 clock cycles to complete each instruction (1 for fetch, 1 for decode, 1 for execute). To process 5 instructions, how many clock cycles are needed?
A pipelined CPU can overlap stages, starting a new instruction each cycle once the pipeline is full. How many cycles would the pipelined CPU need for the same 5 instructions?

○ **A.** Non-pipelined: 15 cycles; Pipelined: 7 cycles

○ **B.** Non-pipelined: 15 cycles; Pipelined: 5 cycles

○ **C.** Non-pipelined: 5 cycles; Pipelined: 3 cycles

○ **D.** Non-pipelined: 8 cycles; Pipelined: 7 cycles

Q26 (3 marks)

A CPU has the following access times:

- Cache access: 5 nanoseconds
- RAM access: 50 nanoseconds

If the cache hit rate is 90% (data is found in cache 90% of the time), what is the average memory access time?

○ **A.** 10 nanoseconds

○ **B.** 27.5 nanoseconds

○ **C.** 45 nanoseconds

○ **D.** 55 nanoseconds

Q27 (3 marks)

A system architect is designing a computer with the following requirements:

- Must address exactly 16 GB of RAM
- Must minimise hardware complexity

How many address lines are needed in the address bus, and what is the reasoning?

○ **A.** 16 address lines, because there are 16 gigabytes

○ **B.** 32 address lines, because 2^{32} = 4 GB and you need extra for headroom

○ **C.** 64 address lines, because modern systems use 64-bit addressing

○ **D.** 34 address lines, because 2^{34} = 16 GB exactly

Q28 (3 marks)

A laptop CPU is rated at 3.5 GHz but during intensive tasks, monitoring software shows it running at only 2.8 GHz. The CPU temperature is showing 95°C (the thermal limit is 100°C). What is happening, and why?

○ **A.** The CPU is defective and needs to be replaced

○ **B.** The operating system is limiting speed to conserve battery power

○ **C.** The software is incorrectly reporting the clock speed

○ **D.** The CPU is throttling its clock speed to reduce heat generation and prevent damage

Q29 (3 marks)

A user wants to upgrade their desktop CPU from a 4-core 3.0 GHz model to an 8-core 4.0 GHz model. They have 8 GB of DDR3-1600 RAM and a standard hard drive (HDD).
After the upgrade, they notice performance has improved but not as much as expected. Which factor is most likely limiting the new CPU's potential?

- ○ **A.** The monitor resolution is too low to benefit from the faster CPU
- ○ **B.** The 8-core CPU requires a minimum of 16 GB RAM to function properly
- ○ **C.** The DDR3-1600 RAM is too slow to keep the faster CPU supplied with data
- ○ **D.** The CPU upgrade was installed incorrectly

Q30 (3 marks)

A chip designer must choose between two cache designs for a new mobile processor:

- **Design X:** 4 MB L3 cache, lower power consumption
- **Design Y:** 12 MB L3 cache, higher power consumption

The processor will be used in smartphones where battery life is critical but users also expect smooth performance. Which design considerations should guide the choice?

- ○ **A.** Design Y should be chosen because larger cache always improves performance significantly
- ○ **B.** Neither design is suitable; mobile processors should not have L3 cache
- ○ **C.** The choice depends on typical workloads — if apps have small working sets, Design X offers better battery life with acceptable performance

○ **D.** Design X should be chosen because smartphones don't benefit from cache memory

Total: **55** marks • Time: **30 min**

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